Learning-Accelerated Distributed Economic Dispatch in Vehicle-to-Grid Integrated Microgrids

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***Abstract*—Next-generation smart grids require decentralized optimization frameworks to coordinate diverse energy assets, including renewables, dispatchable generators, and vehicle-to-grid (V2G)-enabled electric vehicles (EVs). A critical challenge lies in reducing the overall communication overhead and computational burden for distributed economic dispatch through learning-accelerated optimization, while addressing uncertainties in EV charging patterns, time-varying communication networks, and battery longevity constraints. This paper proposes a learning-accelerated distributed optimization framework for V2G-integrated microgrids. A novel alternating direction method of multipliers (ADMM) enhanced by deep neural networks, termed learning-accelerated ADMM (LA-ADMM), is developed to predict optimal dual variables, with theoretical convergence under dynamic network conditions rigorously proved. For handling EV uncertainties, the framework incorporates model predictive control (MPC) architecture with dispatch package strategies that minimize computational overhead through adaptive reuse of historical solutions during steady-state operations. Battery lifespan preservation is achieved through tight coupling of operational constraints with a universal constant current-constant voltage (CC-CV) charging protocol. Validation via IEEE 39-bus simulations and Raspberry Pi-based hardware experiments demonstrates 70.9% iteration reduction compared to ADMM-MPC baselines, effective peak shaving/valley filling, and <2% deviation between simulated and physical power profiles.**

***Index Terms*—Distributed economic dispatch, vehicle-to-grid, alternating direction method of multipliers, deep learning, time-varying network.**

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I. Introduction

NEXT-GENERATION smart grids will increasingly rely on decentralized optimization and secure communication frameworks to coordinate diverse energy assets, including renewables (wind/solar), dispatchable generators (coal/gas/nuclear), demand response programs, and distributed storage with vehicle-to-grid (V2G) integration [1], [2]. A critical challenge involves solving economic dispatch problems through distributed computation while addressing uncertainties in electric vehicle (EV) charging patterns. This requires cost-minimized generation scheduling that meets EV charging demand while adhering to microgrid operational constraints, thereby enabling both rapid generation-load equilibrium and preservation of economic objectives. However, implementing such decentralized solutions proves particularly challenging for large-scale power networks with thousands of nodes.

The alternating direction method of multipliers (ADMM) [3] has emerged as a cornerstone for decentralized power system optimization, particularly in economic dispatch [4], by synergizing the numerical stability of augmented Lagrangian methods with the computational efficiency of dual decomposition [5]. Recent extensions demonstrate ADMM's versatility: residential demand response coordination [6], game-theoretic microgrid planning [7], and privacy-preserving energy management frameworks [8] all employ tailored ADMM implementations. While these advancements [6], [8], [9] achieve theoretical convergence guarantees, they predominantly assume ideal communication infrastructures—a premise often violated by real-world network disruptions like node failures or transmission outages. Contemporary research addresses these limitations through various strategies: stochastic graph modeling for lossy transmission systems [10], virtual buffering mechanisms for packet loss mitigation [11], and consensus-based fault tolerance techniques [12]. However, practical ADMM deployment for economic dispatch remains constrained by sensitivity to parameter selection, necessitating meticulous calibration to ensure both numerical stability and algorithmic convergence [13], [14].

As shown in Table I, existing optimization methods can be classified into five distinct categories: model-based dual decomposition, heuristic tuning, end-to-end optimal solution learning, constraint learning, and warm-start learning. The model-based dual decomposition approach guarantees solution optimality but lacks computational efficiency [3]. Heuristic tuning demonstrates rapid execution but cannot ensure optimal

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TABLE I
COMPARISON OF OPTIMIZATION APPROACHES

	Model-based Decomposition	Heuristic Tuning	Optimal Solution Learning	Constraint Learning	Warm-start Learning
Feasibility	✓	✓			✓
Optimality	✓				✓
Rapidity		✓	✓	✓	✓

solutions [15]. Similarly, end-to-end optimal solution learning [16] and constraint learning [17] achieve computational speed at the expense of solution feasibility—a fundamental optimization requirement. By contrast, warm-start learning synergizes the optimality guarantees of model-based methods with the computational efficiency of learning approaches. This hybrid framework predicts convergence points as initialization vectors for subsequent iterative computations, thereby simultaneously ensuring solution optimality while accelerating convergence dynamics [18]. Initial explorations using fully-connected DNNs for economic dispatch problems were demonstrated in [16], while later work [19] developed specialized ensemble architectures to handle non-convex formulations by explicitly modeling load-to-decision relationships. Decentralized implementations extend this paradigm through localized regression models that approximate optimal power outputs from regional measurements [20]. However, inherent neural approximation errors often lead to constraint violations despite penalty-augmented loss functions. Warm-start learning hybridizes data-driven approaches with conventional optimization by using DNNs to generate initialization points for numerical solvers. This methodology combines the computational efficiency of machine learning with the optimal guarantees of model-based optimization. For AC-OPF acceleration, DNN-generated initializations enable 35× solver speed improvements while maintaining optimality gaps below 0.1% [17]. Distributed variants employ recurrent neural networks (RNNs) to predict ADMM dual variables from partial iteration histories, achieving a two-order-of-magnitude convergence acceleration in DC-OPF solutions [21]. Subsequent extensions [18] generalize these principles through consensus parameter learning on network coupling branches in fully decentralized environments. To the authors' knowledge, existing literature contains no implementations of warm-start learning for decentralized economic dispatch in V2G-enabled microgrids. This approach offers distinctive advantages: consensus parameters in distributed economic dispatch physically correspond to actual pricing mechanisms, enabling market transaction predictions grounded in physical interpretability rather than learned approximations of decentralized algorithms, such as search directions or heuristics.

Model predictive control (MPC) frameworks have become prevalent in addressing EV uncertainties within V2G enabled economic dispatch systems. Reference [22] introduces a decentralized architecture integrating ADMM with MPC, accommodating EV plug-and-play operation while optimizing battery charge-discharge efficiency through predictive horizon coordination. Complementing this, [23] develops a distributionally robust MPC formulation that systematically handles joint uncertainties in electricity market pricing and charging demand profiles. However, conventional MPC implementations incur substantial computational costs due to repeated optimization cycles at each time step. To mitigate this limitation, inspired by [24], [25], the proposed dispatch package technique enables the use from prior optimization results when system state variations remain below predetermined thresholds, effectively reducing both communication overhead and computational burden while maintaining operational feasibility.

Building upon the preceding analysis, this work introduces a learning-accelerated distributed economic dispatch framework with integrated V2G capabilities to address known limitations of existing approaches, specifically: (1) the inability to simultaneously guarantee optimality and computational efficiency, and (2) excessive computational demands from per-step computations in MPC frameworks. The proposed architecture is specifically designed to harness the combined advantages of machine learning, distributed optimization methods, and MPC frameworks while circumventing their inherent limitations. The contributions compared to existing distributed economic dispatch for V2G are summarized as follows:

1) To reduce communication burden and computational load in distributed optimization while guaranteeing system feasibility and optimal solution attainment, a learning-accelerated ADMM (LA-ADMM) mechanism is developed for microgrid economic dispatch with V2G integration, with theoretical convergence guarantees under time-varying communication networks.
2) To address EV uncertainty in V2G operations and computational pressure caused by per-step optimization requirements in MPC frameworks, an MPC architecture augmented with a dispatch package strategy is proposed that dynamically reuses prior optimization results during stable system conditions.
3) To address the battery longevity challenge in V2G systems, the integration of a universal constant current-constant voltage (CC-CV) charging protocol is implemented with physics-informed operational constraints. The proposed dispatch framework's efficacy is validated through simulations and experiments implemented on a real-time system.

II. SYSTEM MODELING AND FORMULATION

This section establishes the cyber-physical framework for microgrid operations with time-varying communication networks, integrating generation dispatch and V2G coordination under constraints, as depicted in Figure 1.

A. Network Framework and Formulation

The communication network is characterized by a time-varying network as $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t), \mathbf{W}(t))$ where $\mathcal{V} = \{1, \ldots, n\}$ is the agent set, $\mathcal{E}(t) \subseteq \mathcal{V} \times \mathcal{V}$ specifies active communication links, and $\mathbf{W}(t) = [w_{ij}(t)]$ denotes the stochastic adjacency matrix capturing connection weights.

The Laplacian $L(t) \in \mathbb{R}^{n \times n}$ satisfies:

$$L(t) := \mathrm{diag}(\mathbf{W}(t)\mathbf{1}) - \mathbf{W}(t) \tag{1}$$

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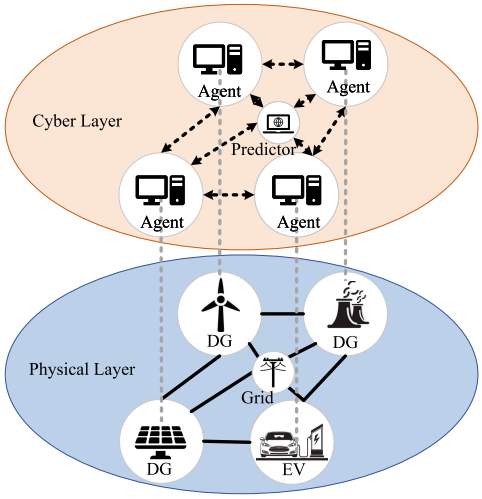

Fig. 1. Cyber-physical framework integrating physical energy assets (DGs, EVs) with cyber-layer predictor and agents.

Each agent $i \in \mathcal{V}$ activates with probability $\alpha_i \in (0, 1]$, while links $(i, j) \in \mathcal{E}(t)$ suffer failures with probability $\beta_{ij} \in [0, 1)$. The effective connectivity matrix $\Gamma(t) = [\gamma_{ij}(t)]$ has entries:

$$\gamma_{ij}(t) = \begin{cases} \alpha_i \alpha_j (1 - \beta_{ij}), & (i, j) \in \mathcal{E}(t) \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

Assumption 1: The network graph $\mathcal{G}(t)$ remains connected $\forall t \geq 0$.

Assumption 1 ensures persistent information diffusion, satisfying standard consensus requirements [26].

B. System Description and Optimization Objective

Consider a Microgrid system comprising $\mathcal{N} = \{1, \ldots, N\}$ dispatchable generators and $\mathcal{M} = \{N+1, \ldots, N+M\}$ electric vehicles over a finite time horizon $\mathcal{T} = \{1, \ldots, \tau\}$. The operational challenge involves minimizing cumulative generation costs while ensuring adequate power supply to meet both static grid demands and dynamic mobility requirements. EV participants arrive stochastically with heterogeneous energy needs and temporal constraints, necessitating coordinated charge-discharge scheduling under finite battery capacities.

The global cost minimization objective is formulated as:

$$\sum_{t \in \mathcal{T}} \left[\sum_{i \in \mathcal{N}} \Phi_i(G_i^t) + \sum_{j \in \mathcal{M}} \Psi_j\left(E_j^t\right) \right] \tag{3}$$

where G_i^t and E_j^t represent power outputs from generator i and net energy exchange from EV j at time t, respectively. The cost functions follow convex quadratic forms:

$$\Phi_i(x) = \alpha_i x^2 + \beta_i x + \gamma_i, \quad \Psi_j(y) = \delta_j y^2 + \epsilon_j y + \zeta_j \tag{4}$$

with strictly positive coefficients $\alpha_i, \delta_j > 0$, and non-negative coefficients $\beta_i, \gamma_i, \epsilon_j, \zeta_j \geq 0$.

Power balance requirements impose:

$$\sum_{i \in \mathcal{N}} G_i^t + \sum_{j \in \mathcal{M}} E_j^t = \sum_{k \in \mathcal{N} \cup \mathcal{M}} D_k^t = D_{sys}^t \quad \forall t \in \mathcal{T} \tag{5}$$

where $D_k^t \geq 0$ denotes the local demand at agent k, with $k \in \mathcal{N}$ representing distributed generators' base loads and $k \in \mathcal{M}$ corresponding to EV charging requirements.

C. Generator Operational Constraints

Each conventional generator $i \in \mathcal{N}$ obeys physical limitations:

$$G_i^{\min} \leq G_i^t \leq G_i^{\max} \quad \forall t \in \mathcal{T} \tag{6}$$

$$-\Delta_i^- \leq G_i^t - G_i^{t-1} \leq \Delta_i^+ \quad \forall t \in \mathcal{T} \tag{7}$$

where $G_i^{\min}$ and $G_i^{\max}$ denote minimum/maximum generation capacities, while Δ_i^- and Δ_i^+ represent downward/upward ramp limits. System adequacy requires:

$$\sum_{i \in \mathcal{N}} G_i^{\min} \leq D_{sys}^t \leq \sum_{i \in \mathcal{N}} G_i^{\max} \quad \forall t \in \mathcal{T} \tag{8}$$

D. Electric Vehicle Dynamics

EV operation incorporates bidirectional energy exchange through V2G interfaces. Let C_j^t and D_j^t denote charging and discharging rates for vehicle $j \in \mathcal{M}$ at time t, constrained by:

$$0 \leq C_j^t \leq \bar{C}_j, \quad 0 \leq D_j^t \leq \bar{D}_j, \quad E_j^t = D_j^t - C_j^t \tag{9}$$

where $\bar{C}_j$ and $\bar{D}_j$ represent maximum charge/discharge rates.

Remark 1: The fundamental complementarity condition prevents concurrent charging and discharging:

$$C_j^t \cdot D_j^t = 0 \quad \forall t \in \mathcal{T}, \forall j \in \mathcal{M} \tag{10}$$

This non-convex constraint ensures mutually exclusive operating modes. However, when both charging efficiency $\theta_j^c < 1$ and discharging efficiency $\theta_j^d < 1$, the inherent energy conversion losses make simultaneous operation suboptimal. Thus constraint (10) can be omitted without affecting solution validity, as proved in [27].

Energy conversion losses are modeled as:

$$\Delta S_j^t = \frac{1}{\theta_j^d} D_j^t - \theta_j^c C_j^t \tag{11}$$

with $\theta_j^c, \theta_j^d \in (0, 1]$ as charge/discharge efficiencies.

Lithium-ion battery states commonly include state-of-charge (SOC), state-of-health, state-of-power, and state-of-energy (SOE). While SOC is frequently employed to represent charge status, its inconsistent measurement units (ampere-hours [Ah] vs. watt-hours [Wh]) create ambiguity. To resolve this ambiguity, the state-of-energy (SOE) is advocated [28], specifically for its consistent energy-based units (Wh) and direct alignment with energy market transactions that occur primarily in watt-hour denominations. The SOE dynamics follow:

$$S_j^t = S_j^{t-1} + \frac{\Delta S_j^t \cdot \Delta t}{\kappa_j} \quad \forall t \in \mathcal{T} \tag{12}$$

where κ_j denotes battery capacity. Temporal constraints for EV j arriving at t_j^{arr} with departure time t_j^{dep} require:

$$\begin{cases} S_j^t \geq S_j^{dep} & t = t_j^{\text{dep}} \\ S^{\min} \leq S_j^t \leq S^{\max} & t_j^{\text{arr}} \leq t \leq t_j^{\text{dep}} \end{cases} \tag{13}$$

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where S_j^{dep} specifies the required departure SOE, and $S^{\min}$, $S^{\max}$ are safety bounds.

The CC-CV charging protocol [29], universally recognized as optimal for lithium-ion batteries, operates through two distinct phases: (1) constant current (CC) charging progressively elevates cell voltage at maximum charging power $\bar{C}_j$ until achieving the voltage at the specific SOE σ_j; (2) constant voltage (CV) maintenance exponentially diminishes current flow to prevent battery degradation. Although computationally simpler, conventional constant power (CP) charging algorithms neglecting CC-CV protocols induce hazardous voltage excursions beyond rated specifications during later charging stages [28], accelerating irreversible battery aging. For tractable optimization implementation, this complex electrochemical behavior is approximated via piecewise linear simplification of empirical data [30], mathematically expressible as:

$$C_j^{t,\max} = \begin{cases} \bar{C}_j, & S_j^t \le \sigma_j \\ \bar{C}_j\left(1 - \frac{S_j^t - \sigma_j}{1-\sigma_j}\right), & S_j^t > \sigma_j \end{cases} \tag{14}$$

where σ_j denotes the CC-CV transition threshold.

III. Proposed Learning-Accelerated Distributed ADMM With MPC Dispatch Packages for V2G

This section presents an integrated optimization framework that combines learning-accelerated ADMM with MPC dispatch packages for V2G systems. By integrating neural network initialization and adaptive time-varying network mechanisms, the proposed method enables efficient energy dispatch while ensuring dynamic coordination in V2G integrated microgrids through MPC driven dispatch package optimization.

A. ADMM With Time-Varying Network Adaptation

The temporal coupling of EV battery dynamics and generator ramp constraints necessitates multi-period coordination. Considering the rolling optimization nature of energy dispatch, let $\omega \in \mathbb{Z}^+$ denote the prediction horizon length. For each decision epoch $t \in \{1, \ldots, \tau-\omega+1\}$, the optimization window is defined as $\mathcal{W}_t = \{t, \ldots, t+\omega-1\}$. The moving-horizon economic dispatch problem becomes:

$$\min_{\substack{G_i^h, E_j^h, \\ C_j^h, D_j^h, \\ S_j^h}} \sum_{h\in\mathcal{W}_t}\left[\sum_{i\in\mathcal{N}} \Phi_i(G_i^h) + \sum_{j\in\mathcal{M}} \Psi_j(E_j^h)\right]$$
$$\text{s.t. Constraints (5)} - \text{(14)}\ \forall h \in \mathcal{W}_t \tag{15}$$

where decision variables G_i^h and E_j^h represent power outputs for generator i and net energy exchange of EV j at prediction step $h \in \mathcal{W}_t$. The local convex feasible sets are characterized by:

- Generators $\mathcal{N}$: G_i^t satisfies (6)-(8)
- EVs $\mathcal{M}$: (C_j^t, D_j^t) satisfies (9)-(14) with $E_j^t = D_j^t - C_j^t$.

From (15), it is observed that constraint (5) represents the sole global constraint, while constraints (6)-(14) constitute local constraints specific to individual agents. This constraint structure permits application of Lagrangian dual decomposition to incorporate the global power balance requirement (5) directly into the objective function through dual variables. Consequently, the optimization problem becomes amenable to distributed parallel computation. Furthermore, the strict convexity of the quadratic cost functions in (4) guarantees zero duality gap [31], enabling the following dual formulation:

$$\min_{\lambda\in\mathbb{R}^\omega} \sum_{k\in\mathcal{N}\cup\mathcal{M}} \Theta_k(\lambda; \mathcal{W}_t) \tag{16}$$

where $\Theta_k(\lambda; \mathcal{W}_t) = \max_{G_k/E_k\in\Omega_k} \sum_{h\in\mathcal{W}_t} -[\Phi_k(G_k^h) + \Psi_k(E_k^h) + \lambda^h(G_k^h + E_k^h - D_k^h)]$.

To enable distributed computation under rolling horizon optimization, window-local dual variables $\lambda_k = [\lambda_k^1, \lambda_k^2, \ldots, \lambda_k^\omega]$ and consensus auxiliary variables z_{kl}^h are introduced for each prediction step $h \in \mathcal{W}_t$. The consensus-constrained dual problem becomes:

$$\min_{\{\lambda_k^h\},\{z_{kl}^h\}} \sum_{k\in\mathcal{N}\cup\mathcal{M}} \sum_{h\in\mathcal{W}_t} \Theta_k^h(\lambda_k^h)$$
$$\text{s.t. } \lambda_k^h = z_{kl}^h,\ \lambda_l^h = z_{kl}^h \quad \forall (k,l) \in \mathcal{E}^h,\ \forall h \in \mathcal{W}_t \tag{17}$$

where $z_{kl}^h \in \mathbb{R}^\omega$ denotes the consensus auxiliary variable for neighbor pair (k, l) at prediction step h; $\Theta_k^h(\lambda_k^h) = \max_{G_k^h/E_k^h\in\Omega_k^h} -[\Phi_k(G_k^h) + \Psi_k(E_k^h) + \lambda_k^h(G_k^h + E_k^h - D_k^h)]$ represents the local dual function; and $\mathcal{E}^h \subseteq (\mathcal{N}\cup\mathcal{M}) \times (\mathcal{N}\cup\mathcal{M})$ describes the time-varying communication topology across prediction steps.

The constraints in (17) enforce neighbor-wise consensus through auxiliary variables z_{kl}^h, ensuring $\lambda_k^h = \lambda_l^h$ for all $(k, l) \in \mathcal{E}^h$. Under Assumption 1, local consensus guarantees global consensus across the prediction window $\mathcal{W}_t$.

The time-varying network challenges are resolved through the formulation:

$$\Sigma_{kl}^{h,(m)} = \begin{cases} \lambda_k^{h,(m)} + \lambda_l^{h,(m)}, & (k,l) \in \epsilon^h \\ \lambda_k^{h,(m-1)} + \lambda_l^{h,(m-1)}, & \{(k,l) \notin \epsilon^h \mid k,l \in \mathcal{V}^h\} \\ 0 & k \notin \mathcal{V}^h \end{cases} \tag{18}$$

$$\Delta_{kl}^{h,(m)} = \begin{cases} \lambda_k^{h,(m)} - \lambda_l^{h,(m)}, & (k,l) \in \epsilon^h \\ \lambda_k^{h,(m-1)} - \lambda_l^{h,(m-1)}, & \{(k,l) \notin \epsilon^h \mid k,l \in \mathcal{V}^h\} \\ 0 & k \notin \mathcal{V}^h \end{cases} \tag{19}$$

where $\epsilon^h \subseteq \mathcal{E}(t)$ denotes active communication links at prediction step h; $\mathcal{V}^h \subseteq \mathcal{N}\cup\mathcal{M}$ represents the set of active agents.

The distributed updates for active agents $k \in \mathcal{V}^h$ become:

$$\gamma_k^{h,(m+1)} = \gamma_k^{h,(m)} + \rho \sum_{l\in\mathcal{N}_k^h} \Delta_{kl}^{h,(m)} \tag{20}$$

$$G_k^{h,(m+1)} = \arg\min_{G_k^h\in\Omega_k^h}\left\{\sum_{h\in\mathcal{W}_t}\left[\Phi_k(G_k^h) + \frac{\rho}{4|\mathcal{N}_k^h|} \times \left\|\frac{1}{\rho}(G_k^h - D_k^h) - \frac{1}{\rho}\gamma_k^{h,(m+1)} + \Sigma_{kl}^{h,(m)}\right\|_2^2\right]\right\} \tag{21}$$

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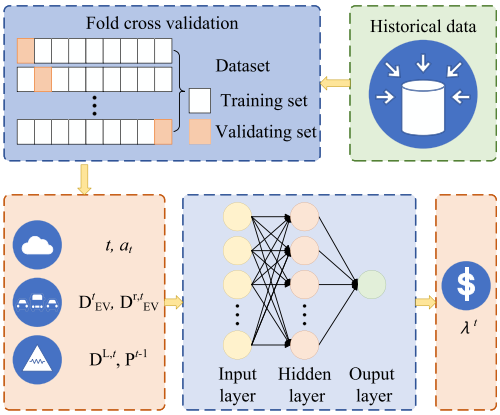

Fig. 2. DNN architecture for learning-accelerated distributed optimization.

$$E_k^{h,(m+1)} = \arg\min_{E_k^h \in \Omega_k^h} \left\{ \sum_{h \in \mathcal{W}_t} \left[\Psi_k(E_k^h) + \frac{\rho}{4|\mathcal{N}_k^h|} \times \left\| \frac{1}{\rho}(E_k^h - D_k^h) - \frac{1}{\rho}\gamma_k^{h,(m+1)} + \Sigma_{kl}^{h,(m)} \right\|_2^2 \right] \right\} \tag{22}$$

$$\lambda_k^{h,(m+1)} = \frac{1}{2|\mathcal{N}_k^h|} \left[\sum_{l \in \mathcal{N}_k^h} \Sigma_{kl}^{h,(m)} - \frac{1}{\rho}\gamma_k^{h,(m+1)} + \frac{1}{\rho}(G_k^{h,(m+1)} + E_k^{h,(m+1)} - D_k^h) \right] \tag{23}$$

Theorem 1: Under Assumption 1, with activation probabilities $\alpha_i \in (0, 1]$ ($\forall i \in \mathcal{V}$) and link failure rates $\beta_{ij} \in [0, 1)$, the distributed ADMM iterations (20)-(23) converge to the optimal primal-dual solutions $\{G_i^{h,*}, E_k^{h,*}\}_{h \in \mathcal{W}_t}$ and $\{\lambda_k^{h,*}\}_{h \in \mathcal{W}_t}$ of Problem (15). The detailed convergence proof under time-varying networks is provided in Appendix A.

B. DNN-Based Dual Variable Prediction

As shown in Figure 2, the proposed DNN accelerates distributed optimization by learning the mapping between operational states and optimal dual variables. Formally, the DNN model $\mathcal{M} : \mathbb{R}^n \to \mathbb{R}^m$ with L hidden layers computes latent representations through hierarchical nonlinear transformations. For input feature vector $\mathbf{x}_t \in \mathbb{R}^n$ at time t, the network computes latent representations $\mathbf{h}_l$ through successive transformations:

$$\begin{aligned} \mathbf{h}_1 &= \mathrm{ReLU}(\mathbf{W}_1\mathbf{x}_t + \mathbf{b}_1) \\ \mathbf{h}_l &= \mathrm{ReLU}(\mathbf{W}_l\mathbf{h}_{l-1} + \mathbf{b}_l), \quad \forall l \in \{2, \ldots, L\} \\ \hat{\lambda}^h &= \mathbf{W}_{L+1}\mathbf{h}_L + \mathbf{b}_{L+1} \end{aligned} \tag{24}$$

where $\mathbf{W}_l \in \mathbb{R}^{d_l \times d_{l-1}}$ and $\mathbf{b}_l \in \mathbb{R}^{d_l}$ denote the weight matrix and bias vector for layer l, with d_l specifying the layer's neuron count. The rectified linear unit (ReLU) activation function $\mathrm{ReLU}(z) = \max(0, z)$ introduces nonlinearity while maintaining gradient stability.

The non-sensitive operational states are encoded in input features $\mathbf{x}^t$:

$$\mathbf{x}^t = \left[t,\ \mathbf{a}^t,\ \mathbf{D}_{\mathrm{EV}}^t,\ \mathbf{D}_{\mathrm{EV}}^{r,t},\ \mathbf{D}^{L,t},\ \mathbf{P}^{t-1} \right]^T \tag{25}$$

where $\mathbf{a}^t \in \{0, 1\}^{N+M}$ is the agent participation vector, $\mathbf{D}_{\mathrm{EV}}^t$ and $\mathbf{D}_{\mathrm{EV}}^{r,t}$ denote the total charging energy demand and residual energy demand for electric vehicles, $\mathbf{D}^{L,t}$ represents the instantaneous power load, and $\mathbf{P}^{t-1}$ stores the power output from the previous time step.

The equilibrium price prediction is explicitly formulated through:

$$\hat{\lambda}^t = f_\theta(\mathbf{x}^t) \tag{26}$$

where $f_\theta(\cdot)$ denotes the pricing model. This predicted price signal serves dual roles as both the operational equilibrium point and the optimal pricing benchmark for microgrid coordination.

Training minimizes a composite loss function over historical trajectories $\mathcal{T}$:

$$\min_{\{\mathbf{W}_l, \mathbf{b}_l\}} \sum_{t \in \mathcal{T}} \underbrace{\left\| \hat{\nu}_t - \bar{\nu}_t \right\|_2^2}_{\text{Prediction error}} + \beta \sum_{l=1}^{L+1} \|\mathbf{W}_l\|_F^2 \tag{27}$$

where the L_2 regularization term with coefficient β prevents overfitting. The dropout layers enhance generalization through stochastic masking:

$$\mathbf{h}_l^{\mathrm{drop}} = \mathbf{h}_l \odot \mathbf{m}_l, \quad \mathbf{m}_l \sim \mathrm{Bernoulli}(0.8) \tag{28}$$

with $\odot$ denoting element-wise multiplication and $\mathbf{m}_l$ masking 20% of neurons during training.

Key Advantages:

1) *Physically Interpretable Pricing*: The dual variable λ^t corresponds to locational marginal pricing in power systems, where the first-order optimality condition:

$$\frac{\partial \Phi_i}{\partial G_i^t} = \frac{\partial \Psi_j}{\partial E_j^t} = \lambda^t \tag{29}$$

holds at equilibrium. This enables direct interpretation of neural predictions as market-clearing prices for microgrid tariff systems.

2) *Privacy-Preserving Feature Design*: Input vectors $\mathbf{x}_t$ exclude sensitive parameters including agent cost functions $\Phi_i(G_i^t)$ and $\Psi_j(E_j^t)$, along with power constraints Ω_i^t, thereby protecting proprietary operational data. All training data originates from real-world operational measurements rather than algorithm-generated iterative data, eliminating the stochasticity inherent in algorithmic iteration processes.

3) *Algorithm Generality*: The methodology generalizes to distributed optimization beyond the ADMM, supporting various power flow formulations including DC optimal power flow and second-order cone relaxations.

The trained DNN provides: (1) Physics-consistent price initialization for accelerated convergence, and (2) Privacy-preserving coordination through non-invasive state observation. This framework extends beyond microgrids to any network requiring distributed resource allocation with market-driven pricing.

C. Dispatch Packet Strategy

To address V2G charging behavior uncertainty, a MPC framework with receding horizon optimization is adopted.

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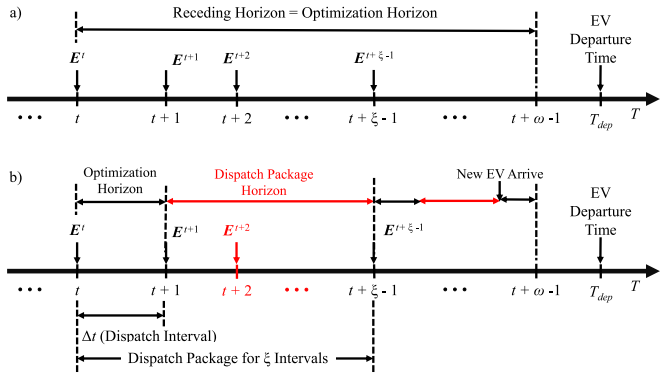

Fig. 3. Temporal structure of a) the convention MPC and b) the proposed dispatch package MPC.

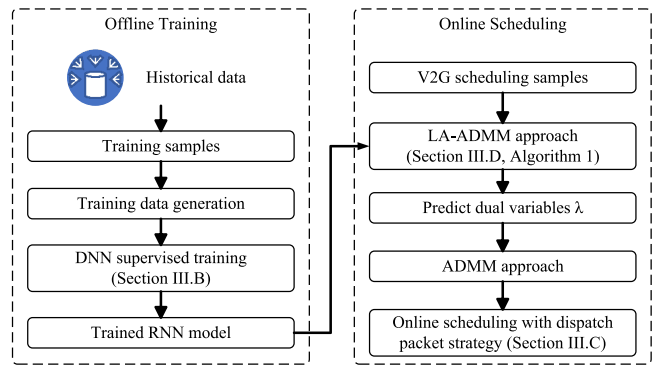

Fig. 4. Integrated offline-online workflow for V2G dispatch scheduling.

The MPC mechanism performs rolling optimization at each timestep to handle stochastic EV participation patterns, where schedule updates occur through continuous prediction and correction cycles. The proposed dispatch packet strategy dynamically adjusts optimization triggering by initiating new computations upon vehicle arrivals and updating schedules upon departures, thereby reducing unnecessary recomputations during stable periods.

The framework's efficiency stems from two complementary components: 1) Standard MPC architecture provides inherent robustness against EV arrival-departure uncertainty through its prediction-correction structure, and 2) The dispatch packet strategy modifies conventional rolling optimization by grouping scheduling signals into multi-step control sequences. As illustrated in Figure 3, each optimized packet spans ξ steps ($\xi \leq \omega$, prediction horizon), with recomputation activated only when new vehicles arrive or at packet expiration boundaries. This combined approach maintains operational stability while significantly reducing computational and communication requirements compared to traditional MPC implementations.

D. Integrated Algorithm

Algorithm 1 details the learning-accelerated ADMM with dispatch package in time-varying networks. The algorithm implements three coordinated phases:

1) **Prediction Phase**: A central RNN predictor synthesizes grid status features to estimate initial dual variables $\hat{\lambda}^t$, significantly reducing subsequent ADMM iterations.

2) **Consensus Optimization**: Agents iteratively update local variables through restricted peer-to-peer communication. The dynamic agent activation mechanism handles topology changes by freezing non-essential agents, ensuring convergence under time-varying networks.

3) **Dispatch Execution**: Optimal solutions are packaged into ξ-interval dispatch commands for sustained operation. Re-computation triggers only when encountering new EVs or reaching update thresholds ($\sigma > \xi$), balancing computational load and control accuracy.

Figure 4 depicts the proposed workflow integrating the LA-ADMM algorithm with the dispatch packet strategy for V2G scheduling. The framework consists of two integrated phases: offline training and online scheduling. In the offline phase, historical data is processed to generate training samples and construct a dataset. This dataset is then used for supervised training of a DNN (Section III-B), employing imitation learning to approximate the equilibrium prices, namely the dual variables λ referenced in (26). During online operation, V2G scheduling samples serve as input. The pre-trained DNN provides high-quality initial estimates for λ, which are used to warm-start the ADMM iterations within the proposed LA-ADMM approach (Algorithm 1, Section III-D), significantly enhancing its convergence efficiency. Finally, the online scheduling process implements the dispatch packet strategy (Section III-C) within an MPC framework.

This architecture integrates learning-based warm-start initialization, distributed consensus optimization, and event-triggered updates, achieving both computational efficiency and optimal operation in dynamic grid environments. However, acquiring high-quality training datasets corresponding to optimal solutions remains time-consuming in real-world implementations.

IV. Experimental Verification

This section validates the efficacy of the proposed learning-accelerated distributed economic dispatch algorithm in accommodating EV plug-and-play operations while ensuring convergence under time-varying communication topologies. The developed methodology not only resolves the optimization problems presented in Section II-B - satisfying both power system energy conservation and EV charging demands—but also achieves communication and computation reduction through accelerated optimization and dispatch package strategies, effectively enabling seamless EV integration with minimized system overhead.

A. Systems Setup and Parameter Configuration

The test system employs an IEEE 39-bus configuration with 10 DGs and 10 EVs, coordinated by 20 cyber agents managing respective DG-EV pairs. This study employs MPC's rolling optimization framework to address uncertainties in EV charging/discharging behaviors, supplemented by a dispatch package strategy designed to reduce communication volume and computational load. Communication uncertainties among

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Algorithm 1 Proposed Learning-Accelerated ADMM With Dispatch Package Over Time-Varying Network

Initialization Initialize system with ξ, ω.
for $t \in \{0, 1, 2, \ldots, \tau\}$ **do**
  if $t = 0$ **or** $\sigma > \xi$ **or** new EV arrivals **then**
    Central predictor receives input features $\mathbf{x}^t = \{t,\ \mathbf{a}^t,\ \mathbf{D}^t_{\mathrm{EV}},\ \mathbf{D}^{r,t}_{\mathrm{EV}},\ \mathbf{D}^{L,t},\ \mathbf{P}^{t-1}\}$
    Predict dual variables $\hat{\lambda}^t \leftarrow \mathrm{RNN}(\mathbf{x}^t)$
    Broadcast $\hat{\lambda}^t$ to all agents. Initialize local variables $\lambda_i^{(0)} = \hat{\lambda}^t$
    while not converged **do**
      for Active agents $i \in \mathcal{V}^h$ (parallel) **do**
        Compute $\Sigma_{kl}^{h,(m)}$ and $\Delta_{kl}^{h,(m)}$ using (18) and (19), respectively;
        Update local variables $\gamma_k^{h,(m+1)}$, $G_k^{h,(m+1)}$, $E_k^{h,(m+1)}$ by solving (20), (21) and (22);
        Update dual variables $\lambda_k^{h,(m+1)}$ using (23);
        Exchange updated $\lambda_k^{h,(m+1)}$ with neighbors $j \in \mathcal{N}_i$
        if stopping criterion is satisfied **then**
        Declare convergence.
        end if
      end for
      for Inactive agents $i \notin \mathcal{V}^h$ (freezing) **do**
        Maintain current states: $\gamma_k^h, G_k^h, E_k^h, \lambda_k^h$
      end for
      Update iteration counter: $m \leftarrow m + 1$
    end while
    Extract the first ξ intervals of dispatch package from the solution (21) and (22);
    Refresh dispatch package counter σ: $\sigma = 1$.
  end if
  Execute dispatch commands. Update counter $\sigma \leftarrow \sigma + 1$
end for

agents during optimization are addressed through algorithmic design. The dynamic evolution of economic dispatch topologies is illustrated in Figure 5, where electric vehicles stochastically engage in bidirectional charging/discharging operations or disconnect from the microgrid. This topological representation indicates that within the MPC rolling framework, connected EVs participate in V2G economic dispatch optimization while disconnected units are excluded from scheduling computations, thereby reducing communication overhead and computational load. It should be noted that economic dispatch topologies remain dynamic prior to optimization commencement but become static once computational processes initiate. Regarding time-varying communication networks emerging during dispatch computations, the algorithm design is detailed in Section III-A. Distributed optimal solutions can be obtained provided the network maintains connectivity during operation. The predictive center solely aggregates agent information and broadcasts predicted dual variables to participating agents, without engaging in actual optimization computations, as shown in Figure 4. Key parameters for DGs and EVs are provided in Tables II and III.

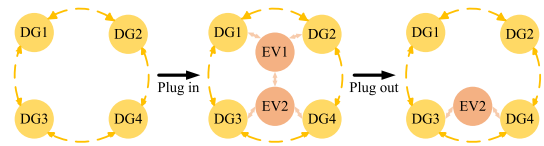

Fig. 5. The evolution of the communication topology when EV1 and EV2 are inserted and disconnected from the microgrid.

TABLE II
PARAMETERS OF DGS [33]

DGi	α_i	β_i	γ_i	$G_i^{\min}(kW)$	$G_i^{\max}(kW)$
0	0.120	3.30	25	10	60
1	0.130	4.50	26	15	60
2	0.125	3.80	28	15	120
3	0.124	3.30	28	13	140
4	0.140	4.50	27	14	200
5	0.118	3.80	26	19	250
6	0.120	3.30	29	18	100
7	0.130	4.50	24	16	110
8	0.125	3.80	23	17	120
9	0.133	3.60	22	15	120

TABLE III
PARAMETERS OF EVS

EVi	δ_j	$\bar{C}_j$	$\bar{D}_j$	θ_j^c	θ_j^d	t_j^{arr}	S_j^{arr}	t_j^{dep}	S_j^{dep}
0	0.05	30	30	0.88	0.88	1	0.4	8	0.6
1	0.05	50	50	0.88	0.88	3	0.4	11	1.0
2	0.02	50	50	0.95	0.95	3	0.5	12	0.8
3	0.04	40	40	0.88	0.88	6	0.8	14	0.4
4	0.03	35	35	0.90	0.90	7	0.2	12	0.8
5	0.06	30	30	0.75	0.75	7	0.4	13	1.0
6	0.06	38	38	0.85	0.85	10	0.5	17	0.7
7	0.03	60	60	0.87	0.87	11	0.8	20	0.4
8	0.06	50	50	0.95	0.95	13	0.5	21	0.8
9	0.03	40	40	0.92	0.92	13	0.8	19	1.0

Temporal constraints enforce SOE safety bounds $S^{\min} = 0$ and $S^{\max} = 1$, with penalty coefficient $\rho = 5$ and convergence threshold $\epsilon = 10^{-4}$. CC-CV charging adheres to the piecewise protocol in (14) with $\sigma_j = 0.85$ [32]. This case study investigates algorithm convergence and optimality under dynamic network conditions with 5% probabilities of agent disconnections and communication failures.

The deep neural network was implemented using TensorFlow 2.8.0 with Keras API in Python 3.11. The model architecture consisted of a sequential network containing two hidden layers (64 and 32 units respectively) with ReLU activation, using L2 weight regularization ($\lambda = 0.001$) and dropout rates of 0.1-0.2 for regularization. The output layer contained a single linear unit for regression prediction. The training protocol employed the Adam optimizer with a base learning rate of 0.001, configured with early stopping (patience=20 epochs) and learning rate reduction on plateau (factor=0.2, patience=10 epochs). Mini-batches of 32 samples were used for gradient updates, with a maximum training duration of 200 epochs monitored through a 20% validation split. The loss function was defined as mean squared error (MSE), with additional metrics including mean absolute error (MAE) and mean absolute percentage error (MAPE).

The synthetic dataset was constructed through parametric variations of hourly load profiles, systematically scaled within [80%, 120%] of nominal demand values. Each modified load

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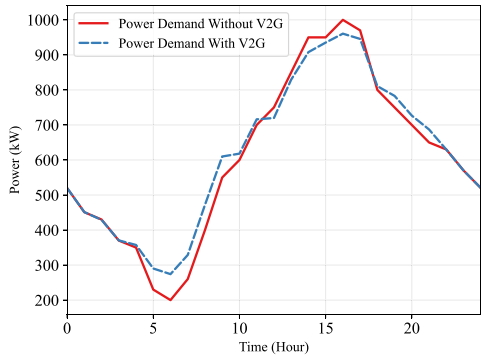

Fig. 6. The total power demand with or without V2G.

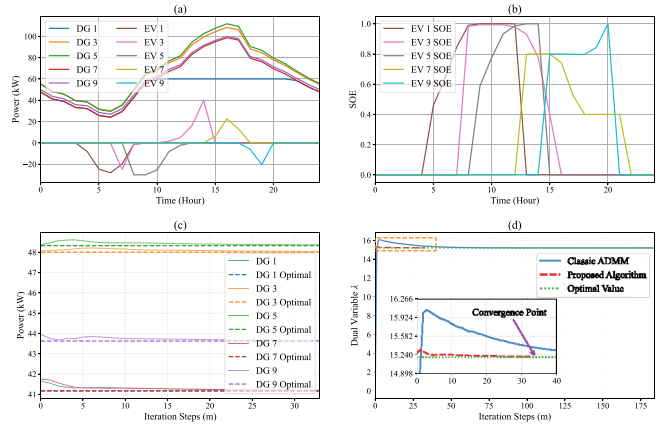

Fig. 7. Performance evaluation of the proposed distributed economic dispatch algorithm in time-varying networks with 10 DGs and 10 EVs (representative trajectories shown for visualization clarity). (a) Optimal 24-hour power schedules for 5 selected DGs and EVs. (b) State of energy (SOE) trajectories of sampled EVs. (c) Convergence to optimal power dispatch at the initial coordination hour ($t = 0$). (d) Dual variable convergence rate comparison: $\lambda_i^h(k)$ of classic ADMM vs. proposed algorithm.

scenario was processed through an ADMM coordination framework with convergence tolerance $\epsilon = 10^{-5}$, generating reference solutions that established the ground-truth labels. These computational results were partitioned using stratified sampling: 80% of instances allocated for model training with 20% of training data reserved for validation monitoring.

B. Performance Comparison Under Non-Ideal Communication

Figure 6 illustrates the power demand profiles under two operational scenarios. The solid red line indicates the baseline demand without V2G integration, while Algorithms 1 coordinate 10 DGs and 10 EVs over a 24-hour horizon to produce the V2G-enabled profile, shown as the dashed blue line and calculated as load minus aggregated EV outputs. This coordinated V2G integration reduces demand variability through effective peak shaving and valley filling, demonstrating significant grid stabilization benefits.

Figure 7(a) displays coordinated 24-hour dispatch plans for 10 DGs and 10 EVs, demonstrating dual-mode operation: peak shaving through strategic discharging during high-demand periods and valley filling via scheduled charging when grid load diminishes. As evidenced in Figure 7(b), all EVs meet their charging requirements, exemplified by EV5 reaching full state of energy (SOE = 1) at hour 11. The characteristic reduced charging slope beyond SOE = 0.85 confirms CC-CV protocol compliance, mitigating battery overvoltage risks and enhancing longevity.

Convergence analysis in Figures 7(c-d) reveals significant improvements: the proposed algorithm achieves optimal dispatch in 34 iterations versus classical ADMM's 184 iterations at initial coordination (t = 0). This acceleration originates from learning-enhanced warm-start initialization near optimal solutions, which reduces both communication frequency and sensitivity to network dynamics. Magnified views demonstrate our method's convergence curve stability compared to classical ADMM's oscillatory patterns under identical network perturbations, confirming enhanced robustness. It should be noted that conventional approaches without dynamic compensation mechanisms fail to converge under such time-varying network conditions.

TABLE IV
ITERATION EFFICIENCY COMPARISON

Algorithm Configuration	Total Iterations
Conventional ADMM + Standard MPC	1,697
ADMM + Proposed Dispatch Package	946
Learning-Accelerated ADMM + Proposed Package	493

C. Computational Efficiency Analysis

The computational advantages of the proposed dispatch package and learning-accelerated ADMM algorithm are validated through 24-hour simulations under time-varying network conditions. As shown in Table IV, three operational configurations are systematically evaluated:

- *Baseline Configuration:* Conventional ADMM with standard MPC rolling window optimization ($\omega = 10$ hours), requiring 1,697 total iterations,
- *Enhanced Package Configuration:* ADMM integrated with the proposed dispatch package (optimized packet span $\xi = 4$), reducing iterations to 946,
- *Complete Solution:* Learning-accelerated ADMM combined with the dispatch package, achieving 493 iterations.

The proposed dispatch package achieves a 44.3% iteration reduction by minimizing optimization frequency under stable system conditions while maintaining the optimized packet span ($\xi = 4$). The combined implementation of learning-accelerated ADMM with the dispatch package achieves 70.9% total reduction compared to the baseline, enabled by neural network-based warm-start initialization that provides near-optimal initial convergence points. This computational efficiency directly translates to proportional reductions in communication overhead and computational load. It should be noted that prediction accuracy does not affect convergence stability but influences convergence speed. Case studies with inaccurate dual variable predictions demonstrate that when prediction errors remain within a bound of 2, the acceleration effect persists; however, when errors exceed this threshold,

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TABLE V
CONVERGENCE PERFORMANCE COMPARISON

Number of iterations / Computation time (s)	50 Agents (40 DGs / 10 EVs)	100 Agents (70 DGs / 30 EVs)	200 Agents (150 DGs / 50 EVs)
Conventional ADMM	115 / 8.66	121 / 19.55	100 / 44.41
Learning-Accelerated ADMM	26 / 2.01	20 / 2.85	23 / 9.91

the acceleration effect diminishes completely, resulting in convergence speeds equivalent to conventional ADMM without learning acceleration.

Comparative analyses between centralized and distributed optimization have been extensively documented in existing literature[13], [34], thus such discussion will not be reiterated herein. This study primarily demonstrates the effectiveness of the proposed neural network-accelerated ADMM algorithm. As shown in Table V, under identical communication networks and convergence conditions, the iteration counts for classical ADMM with 50, 100, and 200 agents are 115, 121, and 100, respectively, while LA-ADMM achieves convergence in 26, 20, and 23 iterations. Corresponding computation times for classical ADMM are 8.66s, 19.55s, and 44.41s, whereas LA-ADMM requires 2.01s, 2.85s, and 9.91s. It is observed that with increasing agents, the iteration counts of classical ADMM exhibit no notable fluctuation, maintaining a range of 100-130 iterations, while the proposed LA-ADMM algorithm maintains convergence within 20-30 iterations, demonstrating significant acceleration of ADMM convergence. Notably, while ADMM serves as the test case, the core methodology—accelerating convergence via neural network-based prediction of convergence points—remains universally applicable to other iterative distributed optimization algorithms. This warm-start approach substantially reduces computation cycles, with acceleration effectiveness being algorithm-agnostic. Furthermore, computation time in distributed optimization correlates with iteration counts and agent numbers, particularly EVs being computationally more intensive than DGs. Nevertheless, the proposed LA-ADMM consistently outperforms classical ADMM in computation efficiency. However, the primary limitation of this methodology lies in the time-consuming process of collecting high-quality solution sets for training neural networks.

D. Hardware Experiment Verification

The learning-accelerated distributed economic dispatch system integrates four operational components: (1) a distributed computing cluster with 8 Raspberry Pi Model B nodes (2GB LPDDR4 RAM, 64-bit quad-core Cortex-A72 processors) for decentralized optimization, (2) a grid-tied bidirectional power device for charge/discharge control, (3) CAN bus signal analysis instrumentation, and (4) four EV battery in a battery test chamber, as shown in Figure 8. The prediction server coordinates initialization parameters through the TCP/IP protocol while all computations execute locally on Raspberry Pi nodes.

Eight Raspberry Pi units represent computational agents, comprising four DG agents and four EV agents. Each Raspberry Pi node operates with an integrated Python-Gurobi academic stack: Python manages distributed communication through socket toolkits while Gurobi handles constrained

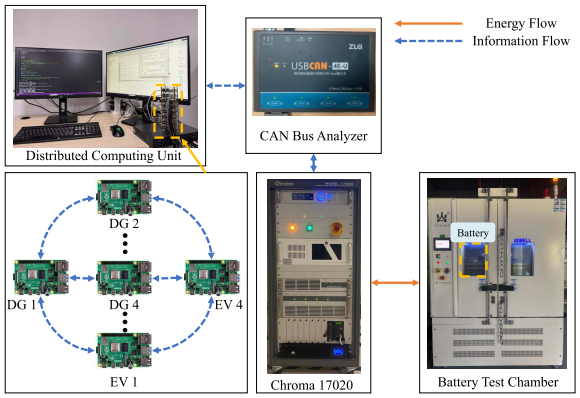

Fig. 8. Experimental platform of distributed economic dispatch.

optimization solving. The computed optimal parameters undergo CAN signal conversion via a dedicated analyzer, executing battery control actions through Chroma 17020 charge/discharge systems. Operational results are streamed to the prediction server, which generates warm-start λ parameters for subsequent optimization cycles using real-time performance data. Updated dual variables are broadcast back to the Raspberry Pi cluster, enabling decentralized ADMM recomputation within a rolling MPC framework. Two battery units at 100% SOE simulate DGs, while two configurable units emulate EVs. Each Raspberry Pi represents one agent, with optimization solutions from two selected DG agents and two EV agents controlling the four batteries. It should be noted that this configuration employs four rather than eight controlled batteries due to hardware limitations (only four battery control channels available on our experiment system). However, the fundamental control logic remains identical for either configuration: optimization signals are transmitted to charging/discharging equipment to execute battery operations. This setup inherently demonstrates the algorithm's scalability. All batteries are housed in a battery test chamber to ensure experimental safety. The hardware-in-the-loop configuration effectively replicates DG-EV microgrid interactions.

Figure 9 experimentally validates the distributed economic dispatch framework through two key metrics: (a) power profile alignment between field measurements and simulations demonstrates $< 2\%$ deviation, while (b) incremental cost converges optimally to 15.184 \$/kWh within 19 iterations by the 3rd operational hour on Raspberry Pi hardware. This empirical evidence confirms both implementation feasibility and accelerated convergence. It should be noted that the real world vs. simulation comparison in Figure 9(a) refers specifically to the alignment between actual charging/discharging power outputs from battery equipment and optimization solutions computed by Raspberry Pi units. Additionally, to effectively demonstrate V2G economic dispatch functionality, this study assumes comparable power scales between DGs and EVs, whereas actual grid operations typically feature DG power capacities substantially exceeding those of individual EVs.

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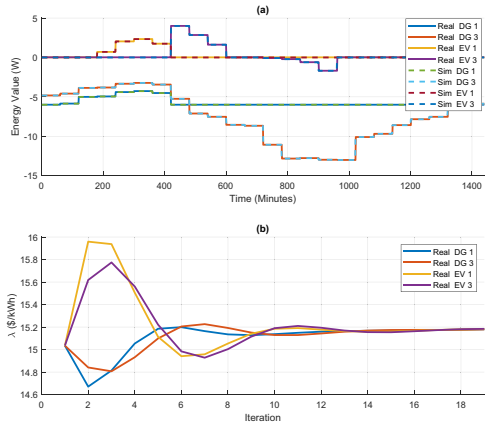

Fig. 9. Experimental result of proposed learning-accelerated distributed economic dispatch algorithm in the time-varying network with compensation. (a) Comparison of real and simulated power over 24h. (b) Convergence of incremental cost iteration of the real world.

Figure 9 experimentally validates the distributed economic dispatch framework through two key metrics: (a) power profile alignment between field measurements and simulations demonstrates $< 2\%$ deviation, while (b) incremental cost converges optimally to 15.184 \$/kWh within 19 iterations by the 3rd operational hour on Raspberry Pi hardware. This empirical evidence confirms both implementation feasibility and accelerated convergence. Due to computational speed and precision limitations of Raspberry Pi hardware, the convergence threshold $\epsilon = 10^{-4}$ was modified to $\epsilon = 10^{-3}$, resulting in fewer convergence iterations in hardware experiments compared to simulation benchmarks. It should be noted that the real world vs. simulation comparison in Figure 9(a) refers specifically to the alignment between actual charging/discharging power outputs from battery equipment and optimization solutions computed by Raspberry Pi units. Additionally, to effectively demonstrate V2G economic dispatch functionality, this study assumes comparable power scales between DGs and EVs, whereas actual grid operations typically feature DG power capacities substantially exceeding those of individual EVs.

V. Conclusion

This study develops a decentralized optimization framework for V2G-enabled microgrids, addressing four challenges: computational and communication pressures arising from prolonged iterations and MPC stepwise calculations, time-varying communication topologies, stochastic EV behaviors, and battery degradation mitigation. The LA-ADMM incorporates embedded deep neural networks to predict dual variable trajectories, achieving 70.9% faster convergence compared to conventional model predictive control-ADMM approaches in 39-bus system simulations. Experimental validation through hardware-in-the-loop tests demonstrates framework robustness, with Raspberry Pi-based physical deployments exhibiting less than 2% power profile deviation from software simulations. This work accelerates V2G economic dispatch through neural network-based initialization while reducing communication and computation loads via a dispatch packet strategy. The incorporation of the CC-CV charging protocol additionally extends battery lifespan. However, the adopted neural architecture lacks domain-specific optimization for energy scheduling dynamics. Future efforts will therefore develop tailored structures embedding both Transformer attention mechanisms and reinforcement learning frameworks. These specialized networks will enhance convergence-point prediction accuracy while preserving operational optimality in smart grid environments.

Appendix A
Proof of Theorem 1

The auxiliary variables z_{kl}^h in (17) facilitate pairwise consensus between neighboring nodes, enforcing $\lambda_k^h = \lambda_l^h$ across all edges $(k, l) \in \mathcal{E}^h$. Assumption 1 ensures these local agreements propagate globally throughout the predictive horizon $\mathcal{W}_t$.

Integrating temporal coordination with decentralized objectives, the augmented Lagrangian formulation becomes:

$$\begin{aligned}\mathcal{L} = \sum_{h\in\mathcal{W}_t}\sum_{k\in\mathcal{N}\cup\mathcal{M}}\Bigg[&\Theta_k^h\left(\lambda_k^h\right) + \sum_{l\in\mathcal{N}_k^h}\Bigg(\mu_{kl}^{h,\top}\left(\lambda_k^h - z_{kl}^h\right)\\ &+ \nu_{kl}^{h,\top}\left(\lambda_l^h - z_{kl}^h\right)\Bigg) + \frac{\rho}{2}\sum_{l\in\mathcal{N}_k^h}\Bigg(\|\lambda_k^h - z_{kl}^h\|_2^2\\ &+ \|\lambda_l^h - z_{kl}^h\|_2^2\Bigg)\Bigg]\end{aligned} \tag{30}$$

where μ_{kl}^h and ν_{kl}^h in $\mathbb{R}^W$ act as dual variables for (5) constraints, with the penalty parameter $\rho > 0$ regulating constraint enforcement. The ADMM procedure iterates through these steps for $m \geq 0$:

$$\mu_{kl}^{h,(m+1)} = \mu_{kl}^{h,(m)} + \frac{\rho}{2}\left(\lambda_k^{h,(m)} - \lambda_l^{h,(m)}\right) \quad \forall (k,l) \in \mathcal{E}^h, \tag{31a}$$

$$\nu_{kl}^{h,(m+1)} = \nu_{kl}^{h,(m)} + \frac{\rho}{2}\left(\lambda_l^{h,(m)} - \lambda_k^{h,(m)}\right) \quad \forall (k,l) \in \mathcal{E}^h, \tag{31b}$$

$$\begin{aligned}\lambda_k^{h,(m+1)} &= \arg\min_{\lambda_k^h}\Bigg\{\sum_{h\in\mathcal{W}_t}\Theta_k^h\left(\lambda_k^h\right) + \lambda_k^h\left(\mu_{kl}^{h,(m+1)} + \nu_{lk}^{h,(m+1)}\right)\\ &\quad + \rho\sum_{l\in\mathcal{N}_k^h}\left\|\lambda_k^h - \frac{1}{2}\left(\lambda_k^{h,(m)} + \lambda_l^{h,(m)}\right)\right\|_2^2\Bigg\}\\ &= \arg\min_{\lambda_k^h}\max_{\substack{G_k\in\Omega_k^h\\ E_k\in\Omega_k^h}}\Bigg\{\sum_{h\in\mathcal{W}_t} -\Phi_k\left(G_k^h\right) - \Psi_k\left(E_k^h\right)\\ &\quad - \lambda_k^h\left(G_k^h + E_k^h - D_k^h\right) + \lambda_k^h\left(\mu_{kl}^{h,(m+1)} + \nu_{lk}^{h,(m+1)}\right)\\ &\quad + \rho\sum_{l\in\mathcal{N}_k^h}\left\|\lambda_k^h - \frac{1}{2}\left(\lambda_k^{h,(m)} + \lambda_l^{h,(m)}\right)\right\|_2^2\Bigg\}\end{aligned} \tag{31c}$$

The min-max structure of subproblem (31c) presents computational challenges. Leveraging the convex-concave duality via the minimax theorem [35] enables transformation to an equivalent max-min form with guaranteed saddle-point solutions:

$$\begin{aligned}\max_{\substack{G_k\in\Omega_k^h\\ E_k\in\Omega_k^h}}\ \min_{\lambda_k^h}\Bigg\{\sum_{h\in\mathcal{W}_t} -\Bigg[&\Phi_k\left(G_k^h\right) + \Psi_k\left(E_k^h\right) + \lambda_k^h\Big(G_k^h + E_k^h\\ &- D_k^h\Big)\Bigg] + \lambda_k^h\left(\mu_{kl}^{h,(m+1)} + \nu_{lk}^{h,(m+1)}\right) + \rho\sum_{l\in\mathcal{N}_k^h}\Big\|\lambda_k^h\end{aligned}$$

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$$-\frac{1}{2}\left(\lambda_k^{h,(m)}+\lambda_l^{h,(m)}\right)\Big\|_2^2\Big\}$$
$$=\max_{\substack{G_k\in\Omega_k^h\\ E_k\in\Omega_k^h}}\min_{\lambda_k^h}\Big\{\sum_{h\in\mathcal{W}_t}\Big[-\Phi_k\left(G_k^h\right)-\Psi_k\left(E_k^h\right)-\frac{\rho}{4|\mathcal{N}_k^h|}$$
$$\Big\|\frac{1}{\rho}\left(G_k^h+E_k^h-D_k^h\right)-\frac{1}{\rho}\left(\mu_{kl}^{h,(m+1)}+\nu_{lk}^{h,(m+1)}\right)$$
$$+\sum_{l\in\mathcal{N}_k^h}\left(\lambda_k^{h,(m)}+\lambda_l^{h,(m)}\right)\Big\|_2^2\Big]+\rho|\mathcal{N}_k^h|\Big\|\lambda_k^h-\frac{1}{2|\mathcal{N}_k^h|}$$
$$\times\Big[\sum_{l\in\mathcal{N}_k^h}\left(\lambda_k^{h,(m)}+\lambda_l^{h,(m)}\right)+\frac{1}{\rho}\Big(G_k^{h,(m+1)}+E_k^{h,(m+1)}$$
$$-D_k^h\Big)-\frac{1}{\rho}\left(\mu_{kl}^{h,(m+1)}+\nu_{lk}^{h,(m+1)}\right)\Big]\Big\|_2^2\Big\}. \tag{32}$$

Quadratic completion establishes this equivalence. Let $G_k^{h,(m+1)}$ and $E_k^{h,(m+1)}$ represent the primal solutions maximizing (31c), forming the saddle-point tuple $(G_k^{h,(m+1)}, E_k^{h,(m+1)}, \lambda_k^{h,(m+1)})$. Consequently, $\lambda_k^{h,(m+1)}$ emerges as the convergent dual solution satisfying (31c)'s optimality criteria.

By substituting (18)-(19) into the update rules and building on [36]'s framework with initialization $\mu_{kl}^{h,(0)}+\nu_{lk}^{h,(0)}=0\ \forall(k,l)\in\mathcal{E}^h$, a simplified iterative scheme is derived:

$$\gamma_k^{h,(m+1)}=\gamma_k^{h,(m)}+\rho\sum_{l\in\mathcal{N}_k^h}\Delta_{kl}^{h,(m)} \tag{33}$$

$$G_k^{h,(m+1)}=\arg\min_{G_k^h\in\Omega_k^h}\Big\{\sum_{h\in\mathcal{W}_t}\Big[\Phi_k\left(G_k^h\right)+\frac{\rho}{4|\mathcal{N}_k^h|}\times\Big\|\frac{1}{\rho}\left(G_k^h-D_k^h\right)-\frac{1}{\rho}\gamma_k^{h,(m+1)}+\Sigma_{kl}^{h,(m)}\Big\|_2^2\Big]\Big\} \tag{34}$$

$$E_k^{h,(m+1)}=\arg\min_{E_k^h\in\Omega_k^h}\Big\{\sum_{h\in\mathcal{W}_t}\Big[\Psi_k\left(E_k^h\right)+\frac{\rho}{4|\mathcal{N}_k^h|}\times\Big\|\frac{1}{\rho}\left(E_k^h-D_k^h\right)-\frac{1}{\rho}\gamma_k^{h,(m+1)}+\Sigma_{kl}^{h,(m)}\Big\|_2^2\Big]\Big\} \tag{35}$$

$$\lambda_k^{h,(m+1)}=\frac{1}{2|\mathcal{N}_k^h|}\Big[\sum_{l\in\mathcal{N}_k^h}\Sigma_{kl}^{h,(m)}-\frac{1}{\rho}\gamma_k^{h,(m+1)}+\frac{1}{\rho}\left(G_k^{h,(m+1)}+E_k^{h,(m+1)}-D_k^h\right)\Big] \tag{36}$$

For Theorem 1's proof, given known values $\{\gamma_k^{h,(m)}, G_k^{h,(m)}, E_k^{h,(m)}, \lambda_k^{h,(m)}\}$, the iteration $m+1$ updates become:

$$\tilde{\gamma}_k^{h,(m+1)}=\gamma_k^{h,(m)}+\rho\sum_{l\in\mathcal{N}_k^h}\Delta_{kl}^{h,(m)}\quad\forall(k,l)\in\epsilon^h, \tag{37}$$

$$\tilde{G}_k^{h,(m+1)}=\arg\min_{G_k\in\Omega_k^h}\Big\{\sum_{h\in\mathcal{W}_t}\Big[\Phi_k\left(G_k^h\right)+\frac{\rho}{4|\mathcal{N}_k^h|}\Big\|\frac{1}{\rho}\left(G_k^h-D_k^h\right)-\frac{1}{\rho}\tilde{\gamma}_k^{h,(m+1)}+\sum_{l\in\mathcal{N}_k^h}\Sigma_{kl}^{h,(m)}\Big\|_2^2\Big]\Big\} \tag{38}$$

$$\tilde{E}_k^{h,(m+1)}=\arg\min_{E_k\in\Omega_k^h}\Big\{\sum_{h\in\mathcal{W}_t}\Big[\Psi_k\left(E_k^h\right)+\frac{\rho}{4|\mathcal{N}_k^h|}\Big\|\frac{1}{\rho}\left(E_k^h-D_k^h\right)-\frac{1}{\rho}\tilde{\gamma}_k^{h,(m+1)}+\sum_{k\in\mathcal{N}_k^h}\Sigma_{kl}^{h,(m)}\Big\|_2^2\Big]\Big\} \tag{39}$$

$$\tilde{\lambda}_k^{h,(m+1)}=\frac{1}{2|\mathcal{N}_k^h|}\Big[\sum_{l\in\mathcal{N}_k^h}\Sigma_{kl}^{h,(m)}-\frac{1}{\rho}\tilde{\gamma}_k^{h,(m+1)}+\frac{1}{\rho}\left(\tilde{G}_k^{h,(m+1)}+\tilde{E}_k^{h,(m+1)}-D_k^h\right)\Big]. \tag{40}$$

Algorithm 1's iterative process satisfies:

$$\tilde{\gamma}_k^{h,(m+1)}=\gamma_k^{h,(m+1)},\quad \tilde{G}_k^{h,(m+1)}=G_k^{h,(m+1)},$$
$$\tilde{E}_k^{h,(m+1)}=E_k^{h,(m+1)},\quad \tilde{\lambda}_k^{h,(m+1)}=\lambda_k^{h,(m+1)}.$$

Leveraging distributed consensus ADMM for the dual problem in (17), [36] establishes the convergence properties, as $m\to\infty$:

$$\tilde{\lambda}_k^{h,(m+1)}\to\lambda^{h,*},\quad \tilde{\lambda}_k^{h,(m+1)}-\tilde{\lambda}_l^{h,(m+1)}\to 0\quad\forall(k,l)\in\mathcal{E}^h. \tag{41}$$

To demonstrate asymptotic optimality, we require:

$$\partial\Phi_k\left(\tilde{G}_k^{h,(m+1)}\right)+\partial\Psi_k\left(\tilde{E}_k^{h,(m+1)}\right)+\lambda^{h,*}\to 0\ \forall k\in\mathcal{N}\cup\mathcal{M}, \tag{42}$$

and fulfills the power balance constraints:

$$\sum_{k\in\mathcal{N}\cup\mathcal{M}}\left(\tilde{G}_k^{h,(m+1)}+\tilde{E}_k^{h,(m+1)}-D_k^h\right)\to 0. \tag{43}$$

The optimality conditions from (34)-(35) yield:

$$0=\sum_{h\in\mathcal{W}_t}\Big(\partial\Phi_k\left(\tilde{G}_k^{h,(m+1)}\right)+\partial\Psi_k\left(\tilde{E}_k^{h,(m+1)}\right)+\frac{1}{2|\mathcal{N}_k^h|}\times\Big[\frac{1}{\rho}\left(\tilde{G}_k^{h,(m+1)}-D_k^h\right)-\frac{1}{\rho}\gamma_k^{h,(m)}+\sum_{l\in\mathcal{N}_k^h}\Sigma_{kl}^{h,(m)}\Big]\Big) \tag{44}$$

Substituting (40) produces the compact form:

$$0=\partial\Phi_k\left(\tilde{G}_k^{h,(m+1)}\right)+\partial\Psi_k\left(\tilde{E}_k^{h,(m+1)}\right)+\tilde{\lambda}_k^{h,(m+1)} \tag{45}$$

As $m\to\infty$, (41) ensures $\tilde{\lambda}_k^{h,(m+1)}\to\lambda^{h,*}$, preserving (45)'s validity.

To prove the power balance (43), rewrite (40) as follows:

$$0=2|\mathcal{N}_k^h|\rho\tilde{\lambda}_k^{h,(m+1)}-\rho\sum_{l\in\mathcal{N}_k^h}\Sigma_{kl}^{h,(m)}+\tilde{\gamma}_k^{h,(m+1)}-\left(\tilde{G}_k^{h,(m+1)}+\tilde{E}_k^{h,(m+1)}-D_k^h\right)$$
$$=2\rho\sum_{l\in\mathcal{N}_k^h}\left(\tilde{\lambda}_k^{h,(m+1)}-\frac{\Sigma_{kl}^{h,(m)}}{2}\right)+\tilde{\gamma}_k^{h,(m+1)}-\Big(\tilde{G}_k^{h,(m+1)}+\tilde{E}_k^{h,(m+1)}-D_k^h\Big)+\rho\sum_{l\in\mathcal{N}_k^h}\left(\Delta_{kl}^{h,(m+1)}-\Delta_{kl}^{h,(m)}\right)$$
$$=\tilde{\gamma}_k^{h,(m+2)}-\left(\tilde{G}_k^{h,(m+1)}+\tilde{E}_k^{h,(m+1)}-D_k^h\right)+\rho\sum_{l\in\mathcal{N}_k^h}\left(\Delta_{kl}^{h,(m+1)}-\Delta_{kl}^{h,(m)}\right) \tag{46}$$

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Aggregating (46) across all devices with $\sum_{k\in\mathcal{N}\cup\mathcal{M}} \gamma_k^{h,(0)} = 0$:

$$\sum_{k\in\mathcal{N}\cup\mathcal{M}} \left(\tilde{G}_k^{h,(m+1)} + \tilde{E}_k^{h,(m+1)} - D_k^h\right) = \rho \sum_{k\in\mathcal{N}\cup\mathcal{M}} \sum_{l\in\mathcal{N}_k^h} \left(\Delta_{kl}^{h,(m+1)} - \Delta_{kl}^{h,(m)}\right). \quad (47)$$

As $m \to \infty$, the dual consensus $\tilde{\lambda}_k^{h,(m+1)} - \tilde{\lambda}_k^{h,(m)} \to 0$ implies $\Delta_{kl}^{h,(m+1)} - \Delta_{kl}^{h,(m)} \to 0$. Substituting into (47) yields:

$$\sum_{k\in\mathcal{N}\cup\mathcal{M}} \left(\tilde{G}_k^{h,(m+1)} + \tilde{E}_k^{h,(m+1)} - D_k^h\right) \to 0. \quad (48)$$

This establishes primal feasibility in (17). Combined with Theorem 1, the algorithm guarantees optimal dispatch in time-varying networks, thus completing the proof.

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